

Question Bank

Unit I: Introduction to Intelligent Transportation Systems (ITS)

1. Explain the historical background of Intelligent Transportation Systems. What were the key milestones in the development of ITS globally and in India?
2. Discuss the challenges and opportunities in the deployment of ITS in urban areas.
3. What are the transport problems and issues that ITS aims to address in the context of urbanization and motorization?
4. Explain the role and importance of ITS in the Indian transport system. Discuss the growth potential for the ITS sector in India.
5. Describe the concept of ITS and its applications. How does ITS contribute to traffic management, public transportation, and vehicle safety?
6. Enumerate the user services offered by ITS. How does each service contribute to the overall functionality of the system?

Unit II: ITS Architecture and Technology

7. Explain the need for ITS architecture in urban traffic management. Describe the components of logical and physical ITS architecture.
8. What are the technology building blocks for ITS? Discuss in detail the data acquisition, communication tools, and data analysis methods used in ITS.
9. What are the various detection and identification methods employed in ITS? Explain their working and applications in traffic monitoring.
10. Explain the concept of Traffic Management Centres (TMC) and their role in the functioning of ITS.
11. List and explain different data collection techniques used in ITS, such as Automatic Vehicle Location (AVL), Automatic Vehicle Identification (AVI), and Geographic Information Systems (GIS).
12. Discuss the importance of telecommunications in ITS. How do communication systems support data fusion in traffic management?
13. With the help of a diagram, explain the physical and logical architecture of an ITS system.
14. Numerical: Calculate the required number of sensors needed for a traffic monitoring system in a city based on the average vehicle count per hour and the maximum detection distance of each sensor.

Unit III: Traffic Management Systems and ITS Applications

15. Discuss the objectives and functions of an Advanced Traffic Management System (ATMS).
16. Explain the role of Advanced Traveller Information Systems (ATIS) in providing real-time traffic data to commuters.
17. Describe the components and working of an Advanced Vehicle Control System (AVCS).
18. What is an Advanced Public Transportation System (APTS)? How does it help in improving urban mobility?
19. Discuss the use of ITS for Commercial Vehicle Operations. How does it improve the efficiency of freight transport?

20. How does ITS contribute to intermodal freight transport?
21. Numerical: A city wants to set up an Advanced Traffic Management System (ATMS) for 10 km of road. The system costs \$500,000/km of road, including installation and operational setup. Calculate the total cost of setting up the system for the city.

Unit IV: ITS Evaluation and Law Enforcement

22. Explain the process of project selection at the planning level for ITS deployment.
23. How is impact assessment carried out for ITS projects? Discuss the key factors that are considered.
24. Discuss the benefits of ITS components in traffic management and law enforcement.
25. What are the funding options available for ITS projects? Explain how governments and private entities can collaborate for ITS funding.
26. Explain how ITS can enhance and support the enforcement of traffic rules and regulations.
27. Discuss the guidelines and procedures for evaluating ITS projects during and after deployment.
28. Numerical: A city has deployed ITS for traffic management at an intersection. The system reduces traffic delays by 15% and increases vehicle throughput by 20%. If the average delay per vehicle before ITS was 5 minutes, calculate the reduction in average delay after ITS implementation.

Unit V: ITS Standards and Smart Cities

29. What is the process of ITS standard development? How do these standards ensure the effective operation of ITS systems?
30. Explain the concept of National ITS Architecture and its application areas in India.
31. Discuss the role of the National Transportation Communications for ITS Protocol (NTCIP) in the functioning of ITS systems.
32. How do ITS standards ensure interoperability across different regions and cities?
33. Discuss the role of ITS in the development of smart cities. What are the key ITS applications in a smart city?
34. Explain the concept of automated highway systems (AHS). How do vehicles in platoons work in such systems?
35. Discuss ITS implementation in developed and developing countries. Provide examples from each.
36. Numerical: Calculate the potential reduction in travel time for a fleet of vehicles when a route guidance system reduces congestion by 25% over a 10 km stretch with an average speed of 40 km/h.

Additional Questions

37. Explain the concept of traffic flow and the fundamental relations of traffic flow in ITS.
38. Discuss traffic stream models and their significance in traffic management.
39. Describe the concept of car-following models. How do they help in simulating and analyzing traffic patterns?

40. What are the components of a lane-changing model, and how do they impact traffic flow in ITS?
41. Discuss the differences between interrupted and uninterrupted traffic flow. How does ITS handle both scenarios?
42. Describe the process of signalized intersection design and analysis based on IRC and HCM standards.
43. Numerical: A signalized intersection has a cycle time of 90 seconds, with green time for the major road being 45 seconds. Calculate the capacity of the intersection if the flow rate is 150 vehicles per hour.
44. What are the key considerations when designing and analyzing traffic signals at intersections?
45. Describe the role of ITS in improving public transportation operations, including real-time tracking and passenger information systems.
46. How does ITS contribute to vehicle safety and reduce accidents?
47. Discuss the role of machine learning and IoT in traffic management. Provide examples of their applications.
48. What are the benefits of electronic toll collection (ETC) systems, and how do they function in ITS?
49. Discuss the role of ITS in sustainable mobility and travel demand management.
50. Numerical: Calculate the total cost savings for a city with an Electronic Toll Collection (ETC) system where the average toll per vehicle is reduced by 30% and the average traffic volume per day is 50,000 vehicles.
51. Traffic Flow and Density Calculations
Given that the traffic flow (Q) is 1200 vehicles per hour, and the average speed of vehicles (V) is 40 km/h, calculate the traffic density (K) in vehicles per kilometer.
52. Capacity of a Lane at an Intersection
A signalized intersection has a cycle time of 90 seconds. The green time for the main road is 50 seconds, and the flow rate is 200 vehicles per hour.
Calculate the capacity of the lane in vehicles per minute.
53. Vehicle Arrival Rate and Queue Length
In a traffic system, the vehicle arrival rate at a signalized intersection is 100 vehicles per hour, and the departure rate is 80 vehicles per hour.
Calculate the average queue length in the system using the formula for queueing theory ($M/M/1$ queue).
54. Impact of ITS on Traffic Delay
A city installs an ITS system that reduces average traffic delays by 20%. If the average delay per vehicle before ITS was 8 minutes, calculate the reduction in delay after ITS installation.
55. Signalized Intersection Performance
The demand for vehicles at a signalized intersection is 1800 vehicles per hour for the main road. The green time provided to the main road is 30 seconds, and the cycle time of the signal is 90 seconds.
Calculate the capacity of the intersection for the main road.
56. Travel Time Savings Due to ITS Deployment
A route length is 10 km. The average speed before the deployment of ITS was 40 km/h. After ITS is deployed, traffic congestion is reduced by 25%, and the average speed increases to 50 km/h.
Calculate the travel time savings due to ITS deployment.
57. Automatic Vehicle Location (AVL) System Coverage
A city has implemented an Automatic Vehicle Location (AVL) system, and the

system covers a network of 200 buses. The AVL system updates the location of each bus every 30 seconds.

Calculate how many data updates the system will generate in one day (24 hours).

58. ITS System Cost vs. Savings Analysis

A city plans to install ITS infrastructure for 50 km of roads. The estimated total cost of installation is \$4 million. After ITS deployment, the system is expected to reduce traffic congestion by 15%, leading to an estimated annual savings of \$300,000 in fuel and time costs.

Calculate the payback period for the ITS system in years.

59. Traffic Flow in a Congested Area

In a congested area, traffic flow is observed to be 1000 vehicles per hour. The density of vehicles in the area is calculated to be 60 vehicles per kilometer.

Calculate the average speed of the vehicles in the area.

60. Fuel Consumption Reduction with ITS

A fleet of commercial vehicles operates with an average fuel consumption of 8 liters per 100 km. After implementing an ITS-based route optimization system, fuel consumption is reduced by 12%.

Calculate the new fuel consumption per 100 km.